

Exercise Sheet 12 – Solutions

Linear Algebra

What do the terms mean?

- **Diagonalizable:** A is called diagonalizable if there is an *invertible* matrix P and a *diagonal* matrix D with

$$D = P^{-1}AP \quad \Longleftrightarrow \quad A = PDP^{-1}.$$

- **What are P and D ?** The **columns of P** are the eigenvectors of A . The **diagonal of D** holds the corresponding eigenvalues – in the *same order* as the columns.
- **Criterion:** An $n \times n$ matrix is diagonalizable $\Leftrightarrow$ there are n linearly independent eigenvectors $\Leftrightarrow$ for *every* eigenvalue $\text{GM} = \text{AM}$ (the eigenspace dimensions sum to n).
- **Reminder:** AM = multiplicity as a root of the char. polynomial; GM = $\dim$ eigenspace = number of independent eigenvectors for that eigenvalue.

[Graphical explanation – not part of the solution]

$$\begin{array}{ccc}
 v \text{ (standard)} & \xrightarrow{\text{blue } A} & Av \text{ (standard)} \\
 \downarrow P^{-1} & & \uparrow P \\
 P^{-1}v \text{ (eigenbasis)} & \xrightarrow[\text{red } D \text{ (just stretch)}]{} & DP^{-1}v \text{ (eigenbasis)}
 \end{array}$$

Diagonalizing means: look at A through the right glasses. Instead of applying A directly, go with P^{-1} into the eigenvector basis, where the map is just a stretch per axis (the diagonal matrix D), then go back with P . That is exactly $A = PDP^{-1}$.

Exercise 47

Problem. Are the matrices diagonalizable over $\mathbb{R}$? If yes, give P (invertible) and D (diagonal) with $D = P^{-1}AP$; if no, explain why.

$$\text{(a)} \quad \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}, \quad \text{(b)} \quad \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}, \quad \text{(c)} \quad \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix}.$$

What needs to be done?

- These are exactly the matrices from **sheet 11** (Exercises 43 and 44). Eigenvalues, AM, GM and eigenspace bases were already computed there – we start with the **summary** (table below).
- Per matrix only check: **is GM = AM everywhere?** If yes → diagonalizable, then P = eigenvectors as columns, D = eigenvalues on the diagonal. If no → not diagonalizable (too few eigenvectors).

Summary from sheet 11.

Matrix	eigenvalue λ	AM, GM	Basis of the eigenspace
(a)	3	2, 2	$(-1, 1, 0), (-1, 0, 1)$
	5	1, 1	$(-1, -2, 1)$
(b)	2	2, 1	$(-1, 0, 1)$
	4	1, 1	$(-1, -1, 1)$
(c)	0	1, 1	$(-5, 0, 1)$
	6	2, 2	$(0, 1, 0), (1, 0, 1)$

[Graphical explanation – not part of the solution]

$$P = \begin{pmatrix} \text{red} & \text{blue} & \text{green} \\ v_1 & v_2 & v_3 \end{pmatrix} \xrightarrow[\text{order}]{\text{same}} D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

Construction: each column of P is an eigenvector, and at the same spot on the diagonal of D sits its eigenvalue. Column 1 belongs to λ_1 , column 2 to λ_2 , etc. – the order must match.

(a) **Matrix** $A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}$

Diagonalizable? Eigenvalues 3 (AM 2, GM 2) and 5 (AM 1, GM 1). For both GM = AM, and $2 + 1 = 3$ independent eigenvectors ⇒ **yes, diagonalizable**.

Build P and D . I take the three eigenvectors as columns (first the two for $\lambda = 3$, then the one for $\lambda = 5$) and write the eigenvalues in the same order onto the diagonal:

$$P = \begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

[Explanation added] Column 1 = $(-1, 1, 0)$ and column 2 = $(-1, 0, 1)$ are the eigenvectors for 3, column 3 = $(-1, -2, 1)$ the one for 5 – that is why the diagonal reads 3, 3, 5.

Check (instead of computing P^{-1}): $D = P^{-1}AP$ is equivalent to $AP = PD$ (just multiply by P from the left). And $AP = PD$ means, column by column: $Av_i = \lambda_i v_i$. That is exactly our eigenvectors:

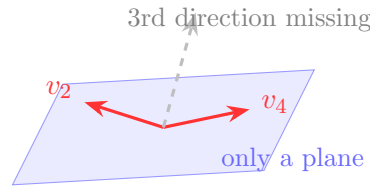
$$A \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = 3 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad A \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}. \quad \checkmark$$

$$A \text{ is diagonalizable with } P = \begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad \text{and } D = P^{-1}AP.$$

(b) **Matrix** $A = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}$

Diagonalizable? Eigenvalues 2 (AM 2, GM 1) and 4 (AM 1, GM 1). For eigenvalue 2 we have GM = 1 < AM = 2.

[Graphical explanation – not part of the solution]



We only have $1 + 1 = 2$ independent eigenvectors (one for 2, one for 4). Two directions span only a plane in $\mathbb{R}^3$ – a third direction for a whole basis is missing. So one cannot build a P from eigenvectors.

Reason. A would be diagonalizable only with 3 independent eigenvectors. But we have only $\text{GM}(2) + \text{GM}(4) = 1 + 1 = 2 < 3$. At eigenvalue 2 an eigenvector “is missing” (GM 1 < AM 2).

A is *not* diagonalizable, because $\text{GM}(2) = 1 < \text{AM}(2) = 2$
(only 2 instead of 3 independent eigenvectors).

(c) **Matrix** $A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix}$

Diagonalizable? Eigenvalues 0 (AM 1, GM 1) and 6 (AM 2, GM 2). Both GM = AM, $1 + 2 = 3$ independent eigenvectors $\Rightarrow$ **yes**.

Build P and D . Columns: the eigenvector for 0, then the two for 6:

$$P = \begin{pmatrix} -5 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix}.$$

Check $AP = PD$, column by column ($Av_i = \lambda_i v_i$):

$$A \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \cdot v_1, \quad A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 6 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 6 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}. \checkmark$$

A is diagonalizable with $P = \begin{pmatrix} -5 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix}$, and $D = P^{-1}AP$.

Exercise 48

Problem. For which values of the parameter a is the matrix diagonalizable?

(a) $A = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}, \quad \text{(b) } A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ a & 2a & 3a \end{pmatrix}.$

(a) $A = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}$

What needs to be done?

- Set up the char. polynomial (depends on a), read off the eigenvalues and their AM, then check the GM of the repeated eigenvalue. Diagonalizable $\Leftrightarrow$ GM = AM everywhere.

Step 1: Char. polynomial. On the diagonal of $A - tI$ sits $a - t$; I abbreviate $s := a - t$. Then

$$\det(A - tI) = \det \begin{pmatrix} s & 1 & 1 \\ 1 & s & 1 \\ 1 & 1 & s \end{pmatrix}.$$

With the Sarrus rule ($\det = aei + bfg + cdh - ceg - afh - bdi$, here all off-diagonal entries = 1):

$$= \underbrace{s^3}_{aei} + \underbrace{1}_{bfg} + \underbrace{1}_{cdh} - \underbrace{s}_{ceg} - \underbrace{s}_{afh} - \underbrace{s}_{bdi} = s^3 - 3s + 2.$$

Factor: $s = 1$ is a root ($1 - 3 + 2 = 0$), splitting it off gives $s^3 - 3s + 2 = (s - 1)^2(s + 2)$.

Step 2: Eigenvalues (back with $s = a - t$, i.e. $t = a - s$). The roots $s = 1$ (double) and $s = -2$ (simple) become

$$t = a - 1 \text{ (AM 2)}, \quad t = a + 2 \text{ (AM 1)}.$$

These are *always* different ($a - 1 \neq a + 2$), no matter which a .

Step 3: GM of the double eigenvalue $a - 1$. Here is the nice part – I subtract $(a - 1)I$:

$$A - (a - 1)I = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix} - \begin{pmatrix} a - 1 & 0 & 0 \\ 0 & a - 1 & 0 \\ 0 & 0 & a - 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

That is the “all-ones” matrix – all rows equal, so **rank** 1. Hence

$$\text{GM}(a - 1) = \dim \ker = 3 - 1 = 2 = \text{AM}(a - 1). \quad \checkmark$$

The simple eigenvalue $a + 2$ automatically has $\text{GM} = 1 = \text{AM}$. **This holds for every a** – the matrix $A - (a - 1)I$ is *always* the all-ones matrix.

A is **diagonalizable for all $a \in \mathbb{R}$.**

[Explanation added] No surprise: A is symmetric ($A = A^\top$), and real symmetric matrices are always diagonalizable. Our route shows it directly via $\text{GM} = \text{AM}$.

(b) $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ a & 2a & 3a \end{pmatrix}$

[Graphical explanation – not part of the solution]

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ a & 2a & 3a \end{pmatrix} \left\{ \begin{array}{l} \text{row 1} = 1 \cdot (1, 2, 3) \\ \text{row 2} = 2 \cdot (1, 2, 3) \\ \text{row 3} = a \cdot (1, 2, 3) \end{array} \right.$$

Every row is a multiple of $(1, 2, 3)$: the 1-, 2- resp. a -fold. Such a matrix has **rank** 1 (all rows “on one line”). Just like the rank-1 matrix on sheet 11 (44b), almost all eigenvalues are then 0.

Step 1: Rank and eigenvalue 0. All rows are multiples of $(1, 2, 3) \Rightarrow \text{rank}(A) = 1$. Dimension formula: $\dim \ker(A) = 3 - 1 = 2$. So 0 is an eigenvalue with GM(0) = 2.

Step 2: the other eigenvalue. A rank-1 matrix has only *one* eigenvalue $\neq 0$, and it equals the trace:

$$\text{trace}(A) = 1 + 4 + 3a = 5 + 3a.$$

The char. polynomial is $\chi_A(t) = t^2(t - (5 + 3a))$ – roots 0 (double) and $5 + 3a$.

Step 3: Case distinction.

- $5 + 3a \neq 0$, i.e. $a \neq -\frac{5}{3}$: eigenvalues 0 (AM 2, GM 2) and $5 + 3a$ (AM 1, GM 1). GM = AM everywhere $\Rightarrow$ **diagonalizable**.
- $5 + 3a = 0$, i.e. $a = -\frac{5}{3}$: now the third eigenvalue is 0 too, i.e. 0 has AM 3. But GM(0) = 2 (the rank stays 1). Since GM(0) = 2 < 3 = AM(0) **not diagonalizable**.

A is diagonalizable **exactly when** $a \neq -\frac{5}{3}$.
For $a = -\frac{5}{3}$, A is *not* diagonalizable (GM(0) = 2 < AM(0) = 3).

Exercise 49

Problem. The sequence $(a_0, a_1, a_2, \dots)$ is given by

$$a_0 = 4, \quad a_1 = -1, \quad a_k = 3a_{k-1} + 10a_{k-2} \quad (k \geq 2).$$

Find an explicit formula for a_k using **matrix diagonalization**.

What needs to be done?

- Pack two consecutive terms into a vector – then “one step of the sequence” becomes *one* matrix multiplication $v_k = Mv_{k-1}$. So $v_k = M^k v_0$. The high power M^k is computed via $M = PDP^{-1}$, because $M^k = PD^kP^{-1}$ and D^k is just the k -th power of the diagonal.

[Graphical explanation – not part of the solution]

$$\begin{array}{c} a_k = 3a_{k-1} \\ + 10a_{k-2} \end{array} \xrightarrow{\text{pack}} v_k = \begin{pmatrix} a_{k+1} \\ a_k \end{pmatrix} \xrightarrow{M^k = PD^kP^{-1}} v_k = M^k v_0$$

The sequence “looks two steps back”. Packing (a_{k+1}, a_k) into a vector, the matrix M turns the old pair into the new one. Applying it k times = M^k – and that is easy via diagonalization, because D^k is just the entries to the power k .

Step 1: Vector and matrix. Set $v_k = \begin{pmatrix} a_{k+1} \\ a_k \end{pmatrix}$. With $M = \begin{pmatrix} 3 & 10 \\ 1 & 0 \end{pmatrix}$ we have

$$Mv_{k-1} = \begin{pmatrix} 3 & 10 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_k \\ a_{k-1} \end{pmatrix} = \begin{pmatrix} 3a_k + 10a_{k-1} \\ a_k \end{pmatrix} = \begin{pmatrix} a_{k+1} \\ a_k \end{pmatrix} = v_k.$$

Start vector $v_0 = \begin{pmatrix} a_1 \\ a_0 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, and thus $v_k = M^k v_0$.

Step 2: Diagonalize M . Char. polynomial:

$$\det(M - tI) = \det \begin{pmatrix} 3-t & 10 \\ 1 & -t \end{pmatrix} = (3-t)(-t) - 10 = t^2 - 3t - 10 = (t-5)(t+2).$$

Eigenvalues $\lambda = 5$ and $\lambda = -2$. Eigenvectors:

- $\lambda = 5$: $M - 5I = \begin{pmatrix} -2 & 10 \\ 1 & -5 \end{pmatrix}$, row 1: $-2x + 10y = 0 \Rightarrow x = 5y$. eigenvector $(5, 1)$.
- $\lambda = -2$: $M + 2I = \begin{pmatrix} 5 & 10 \\ 1 & 2 \end{pmatrix}$, row 2: $x + 2y = 0 \Rightarrow x = -2y$. eigenvector $(-2, 1)$.

$$P = \begin{pmatrix} 5 & -2 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}, \quad \det P = 7, \quad P^{-1} = \frac{1}{7} \begin{pmatrix} 1 & 2 \\ -1 & 5 \end{pmatrix}.$$

Step 3: $v_k = PD^kP^{-1}v_0$. Compute right to left. First $P^{-1}v_0$:

$$P^{-1}v_0 = \frac{1}{7} \begin{pmatrix} 1 & 2 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} -1+8 \\ 1+20 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 7 \\ 21 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Then D^k in front ($D^k = \begin{pmatrix} 5^k & 0 \\ 0 & (-2)^k \end{pmatrix}$):

$$D^k P^{-1}v_0 = \begin{pmatrix} 5^k & 0 \\ 0 & (-2)^k \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5^k \\ 3(-2)^k \end{pmatrix}.$$

Finally P in front:

$$v_k = \begin{pmatrix} 5 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 5^k \\ 3(-2)^k \end{pmatrix} = \begin{pmatrix} 5 \cdot 5^k - 6(-2)^k \\ 5^k + 3(-2)^k \end{pmatrix}.$$

Step 4: Read off a_k . a_k is the *lower* component of v_k :

$$a_k = 5^k + 3 \cdot (-2)^k.$$

Check: $a_0 = 1+3 = 4 \checkmark$, $a_1 = 5+3 \cdot (-2) = -1 \checkmark$, $a_2 = 25+3 \cdot 4 = 37 = 3(-1)+10(4) \checkmark$.

Overview: Results

Exercise	Result
47 (a)	diagonalizable, $D = \text{diag}(3, 3, 5)$
47 (b)	<i>not</i> diagonalizable ($\text{GM}(2) = 1 < \text{AM}(2) = 2$)
47 (c)	diagonalizable, $D = \text{diag}(0, 6, 6)$
48 (a)	diagonalizable for all $a \in \mathbb{R}$
48 (b)	diagonalizable $\Leftrightarrow a \neq -\frac{5}{3}$
49	$a_k = 5^k + 3 \cdot (-2)^k$